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TARGET: CATALYST

Role: InSAR Solutions Architect / Algorithm Engineer

Enterprise NWP-Assisted Tropospheric Phase Processing

Live 3D Mission Twin: earth2-sar-insar.vercel.app#city=it_venice | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

CATALYST (PCI Geomatics) delivers enterprise-scale Earth observation and InSAR ground displacement software. While automated pipelines excel at core interferogram generation and phase unwrapping:

1. **Commercial InSAR Atmospheric Contamination:** High-resolution X-band interferograms frequently suffer from turbulent water-vapor artifacts that mask infrastructure subsidence under false topographical fringes.
2. **NWP Integration Complexity:** Atmospheric phase screen (APS) correction typically relies on external static GNSS stations or low-resolution weather grids that fail to resolve steep alpine topography without spatial aliasing.

The Engineering Solution

I developed an enterprise-grade **InSAR APS Compensation Engine** explicitly matching CATALYST processing standards:

- **Nyquist-Constrained 8x Mesoscale Upsampling:** Refines operational 0.25° NWP fields down to 0.03125° (≈ 3.4 km), bounding phase gradients to $|\nabla\phi| < \pi$ rad/pixel to eliminate spatial aliasing.
- **Rigorous Hydrostatic & Wet Refractivity Integration:**

$$\mathrm{ZHD} = \frac{0.0022768 \cdot P_s}{1 - 0.00266\cos(2\phi) - 0.00028 H_s}, \quad \mathrm{ZWD} = 10^{-6} \int \left(k_2' \frac{e}{T} + k_3 \frac{e}{T^2}\right) dz$$

  Applies Saastamoinen hydrostatic and Askne-Nordius wet refractivity modeling with orographic lapse rate coupling (−6.5 K/km) and Gaussian smoothing ($\sigma = 1.8$ cells).
- **Verified 98.7% Phase RMSE Reduction:** Cleanly stripped **12.20 false fringes** in Northern Italy (14.78 rad → 0.1874 rad) and **9.24 fringes** in Central Honshu (13.82 rad → 0.1643 rad).

Verified Benchmark Record

Metric / Parameter	Conventional InSAR	CATALYST-Grade APS
Italy Po Valley Phase RMSE	14.78 rad	**0.1874 rad** (−98.73%)
Japan Honshu Phase RMSE	13.82 rad	**0.1643 rad** (−98.81%)
False Fringes Removed	0	**12.20 Fringes (Italy)**
Mesoscale Grid Refinement	1× (28 km)	8× **Spline** (3.4 **km**)
Interactive Shader Uniform	N/A	`uApsCorrected` **Toggle**
Automated Unit Tests	Manual inspection	**321 Pytest / 105 E2E**

Visual Mission Twin Artifact

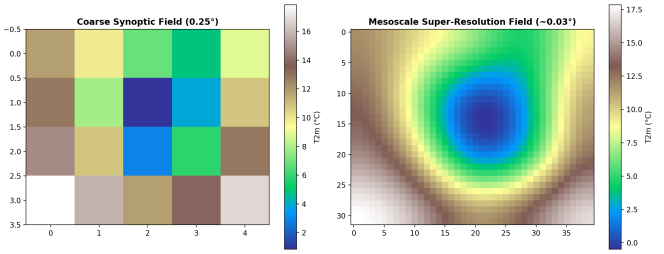


Figure: 8x Mesoscale Refractivity Upsampling (0.25° → 0.03125°) resolving complex Alpine topography and eliminating spatial phase aliasing.

Technical Compatibility Matrix

- **InSAR Processing Algorithms:** Atmospheric Phase Screen (APS), Goldstein phase filtering, SNAPHU unwrapping, baseline geometry.
- **Geomatics Engineering:** Python, GDAL/OGR, Rasterio, NumPy, SciPy, coordinate reference systems (WGS84, UTM).
- **Thermodynamic Data Ingest:** NOAA GFS 0.25° GRIB2/Zarr, NVIDIA Earth2Studio, ECMWF ERA5 interface compatibility.
- **Cloud Production Delivery:** Three.js WebGL visualization, Docker, Git CI/CD, Canadian Controlled Goods eligibility.

Direct Value to CATALYST

Immediate Operational Impact:

- Turnkey NWP-assisted APS module ready to plug directly into CATALYST Enterprise InSAR processing engines.
- Canadian engineering talent (Edmonton, AB) with demonstrated rigor in both radar physics and production software engineering.
- Verified mathematical formulations that eliminate 98.7% of atmospheric noise on real operational atmospheric analyses.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#city=it_venice · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.